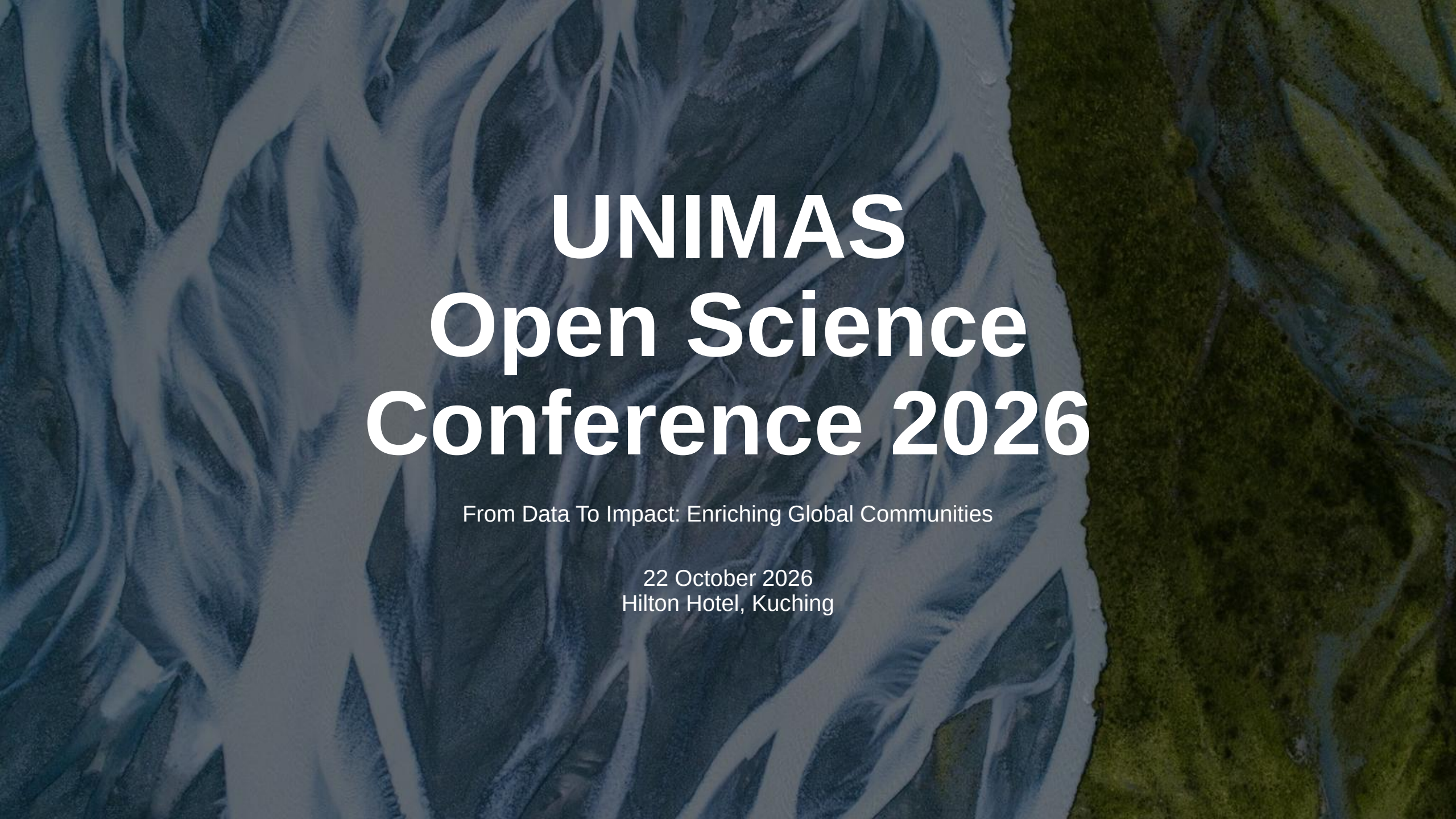


UNIMAS Open Science Conference 2026

From Data To Impact: Enriching Global Communities

22 October 2026
Hilton Hotel, Kuching

Team Members

- Dr Jongkar Grinang
- Associate Professor Dr Samuel Lihan
- Professor Gabriel Tonga Noweg
- Sabella Justin
- Selvana Felix
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- Mazzaellynn Brasenia Umang Thomas
- Francis Senen Alau

- In memory of TR Kedit Bedin of Raba Tiput, Upper Layar, Betong





Introduction

- Defining citizen science?
  - Public participation in scientific research
  - Collaboration between professional scientists and local volunteers
  - Volunteers or citizen scientists involved in co-designing research approaches, collecting and categorizing field data
  - Analyzing and communicating findings
- Types of citizen science:
  - Contributory
  - Collaborative
  - Co-creation

Research

- Citizen Science Approach in Recording Species Occurrence of Fish and Crustaceans in the Upper Layar River, Betong, Sarawak
- Citizen Science Approach Among Indigenous Communities in Assessing Biological Diversity and Environmental Quality in the Upper Layar Area, Betong, Sarawak
- Citizen Science Approach in Assessing Harvest Intensity of Ethnobotanical Resources Among Indigenous Communities in the Upper Layar Area, Betong, Sarawak



Location

- Iban longhouse communities in Upper Layar River area, Betong Division, Sarawak
  - Raba Tiput
  - Nanga Sela
  - Nanga Selangking
  - Nanga Mara
  - Tembawai Tinting



Participants

- Participants involved in planning and trained in different scientific field methods (esp. sample collection)
    - Active discussions on local knowledge and its complexities
- Longhouse community members:
    - Gender
    - Youths
    - Status
- Activities include forest transect, collection of specimens, identifying local taxonomies, sampling, interviews, GPS mapping







UNIMAS
UNIVERSITI MALAYSIA SARAWAK

INSTITUTE OF BIODIVERSITY & ENVIRONMENTAL CONSERVATION

UNIMAS

Open Science Conference

"From Data to Impact: Enriching Global Communities"

22nd October 2026

Hilton Kuching, Sarawak

Authors:

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From *Enseriban* to *Rasbora sarawakensis*

Bridging scientific taxonomy and Iban fish nomenclature through citizen science in Ulu Layar, Sarawak, Borneo



Two knowledge systems, one personal journey





> 50 years ago

- Iban community knowledge
- Grandparents • parents • elders • rivers • fish names

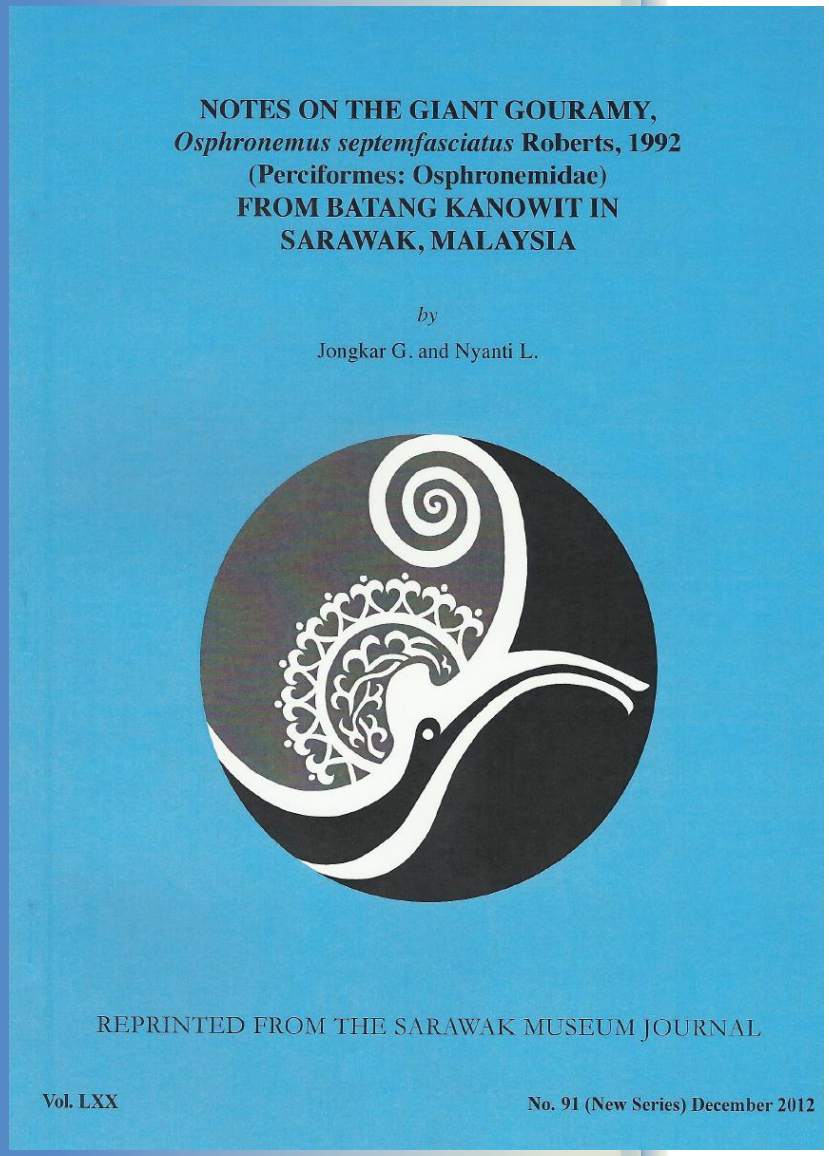
> 30 years ago

- Scientific taxonomy
- Specimens • morphology • nomenclature • classification

Today

- Citizen science
- Bringing the two knowledge systems into dialogue

The central question


NOTES ON THE GIANT GOURAMY,
Osphronemus septemfasciatus Roberts, 1992
(Perciformes: Osphronemidae)
FROM BATANG KANOWIT IN
SARAWAK, MALAYSIA

by

Jongkar G. and Nyanti L.

REPRINTED FROM THE SARAWAK MUSEUM JOURNAL

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Does Iban fish nomenclature represent an underlying folk taxonomy?

Scientific taxonomy
ICZN

Formal identification, diagnosis, classification and scientific nomenclature.

↔

Iban fish nomenclature
LEK

Local names that may encode recognition, grouping and biological distinctions.

↑

Citizen science

The collaborative space where nomenclature can be compared with scientific taxonomy and its underlying classification explored.

From Iban nomenclature to folk-taxonomy hypothesis

Iban Indigenous Ecological Knowledge
Long-term interaction with river and fishes

→

Iban Fish Nomenclature
Enseluai • Enseriban • Belau • Runtu

→

Citizen Science
Participation • survey • documentation

→

Scientific Taxonomic Validation
Genus- and species-level identification

→

Compare
Correspondence

Observed nomenclatural–taxonomic correspondence

Species-level correspondence
Enseriban ↔ *Rasbora sarawakensis*
Entabalang ↔ *Barbodes kuchingensis*
Runtu ↔ *Channa lucius*

Broader correspondence
Enseluai ↔ several *Rasbora* spp.
Belau ↔ several *Channa* spp.
Baung ↔ several *Hemibagrus* spp.

Does Iban fish nomenclature reflect an underlying folk taxonomy?
A hypothesis suggested by correspondence patterns—not yet fully demonstrated

Why Ulu Layar?

- **A remote river landscape where biodiversity knowledge is distributed across people and place.**

Scientific challenge

- Remote freshwater diversity remains incompletely documented.
- Access, cost and limited taxonomic expertise constrain conventional surveys.
- Small-bodied or uncommon species remain difficult to detect.

Community knowledge

- Fishing knowledge accumulated across generations.
- Familiarity with habitats, seasons and fishing methods.
- Fish names embedded in everyday resource use.

Open Science opportunity

- Community participation expands observation.
- Scientific validation makes records reusable.
- Knowledge exchange works in both directions.

ULU LAYAR AS A LIVING LABORATORY







The Ulu Layar case study

Duration

13 months
Dec 2023 – Dec 2024

Community context

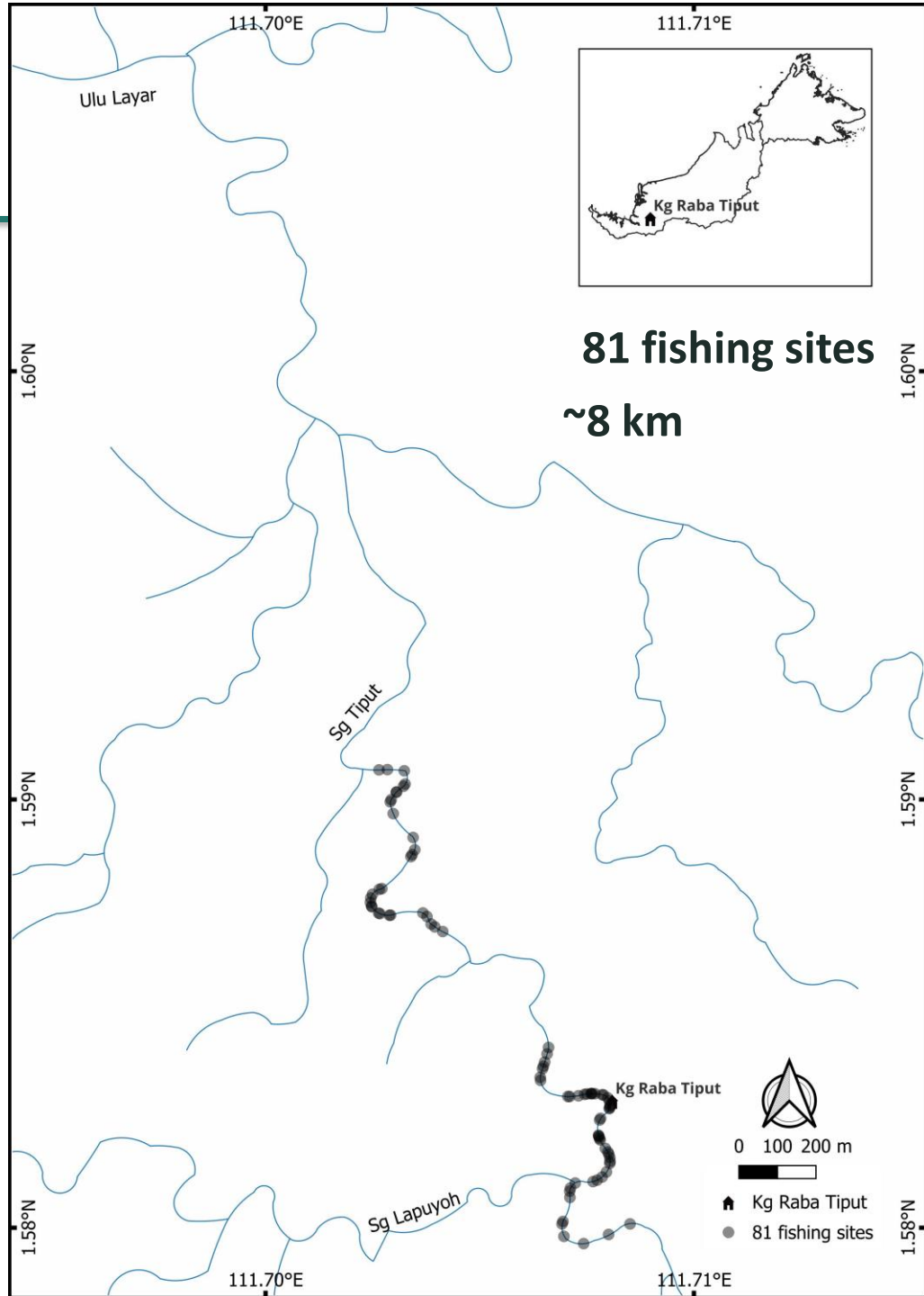
Traditional harvesting + local river knowledge

Conservation context

Tagang no-fishing zones



Raba Tiput Iban community



Citizen science workflow

- From community knowledge to taxonomically validated biodiversity records

1

Planning

Community selection, prior informed consent, shared objectives

2

Training

Standard protocols, equipment, fish identification, recording

3

Implementation

Volunteers independently survey and submit fish records

4

Documentation

Taxonomic verification, validation and analysis







How do we test for an underlying folk taxonomy?

Genus-rank correspondence

G-Yes

A vernacular name consistently refers to fishes belonging to the same scientific genus, regardless of species identity.

Species-rank correspondence

S-Yes

A vernacular name uniquely corresponds to one scientific species.

Broader folk taxon

S-No

A name may include multiple scientific species, a broader group, or distinguish a life stage.

Key logic:
Repeated correspondence, selective species-level naming and coherent broader groupings may indicate an underlying folk classification—not simply naming accuracy.



Kejampa - *Glyptothorax*



Enseluai - *Rasbora*

First evidence: strong taxonomic correspondence

100%

Genus-level correspondence

≈60%

Unique species-level correspondence

This pattern raises the hypothesis that Iban nomenclature reflects an underlying biologically coherent folk taxonomy—not merely a collection of informal common names.

Where the two systems align one-to-one

- Examples of species-specific Iban names in Ulu Layar

Bangah

Barbodes banksi



Entabalang

Barbodes kuchingensis



Kepiat

Barbonymus collingwoodii



One-to-one naming suggests consistent recognition of these species.

The diagnostic cues used by fishers remain to be investigated directly.

The key case: Enseluai, Enseriban and *Rasbora*

- Different naming resolution can reveal how fishes are locally classified

ENSELUAI

Used for several species of *Rasbora*

Scientific system: separates multiple species
Iban system: groups morphologically similar members under a broader folk taxon



Rasbora tubbi



Rasbora einthovenii

ENSERIBAN

Used specifically for ***Rasbora sarawakensis***

A distinct local name is retained for one species within the broader *Rasbora* assemblage.



This nested pattern is important evidence to investigate as a possible folk-taxonomic structure.

A second clue: Belau and Runtu





SCIENTIFIC TAXONOMY		IBAN NOMENCLATURE
Channa spp.		Belau
Channa gachua	→	Belau
Channa striata	→	Belau
Channa lucius	→	Runtu

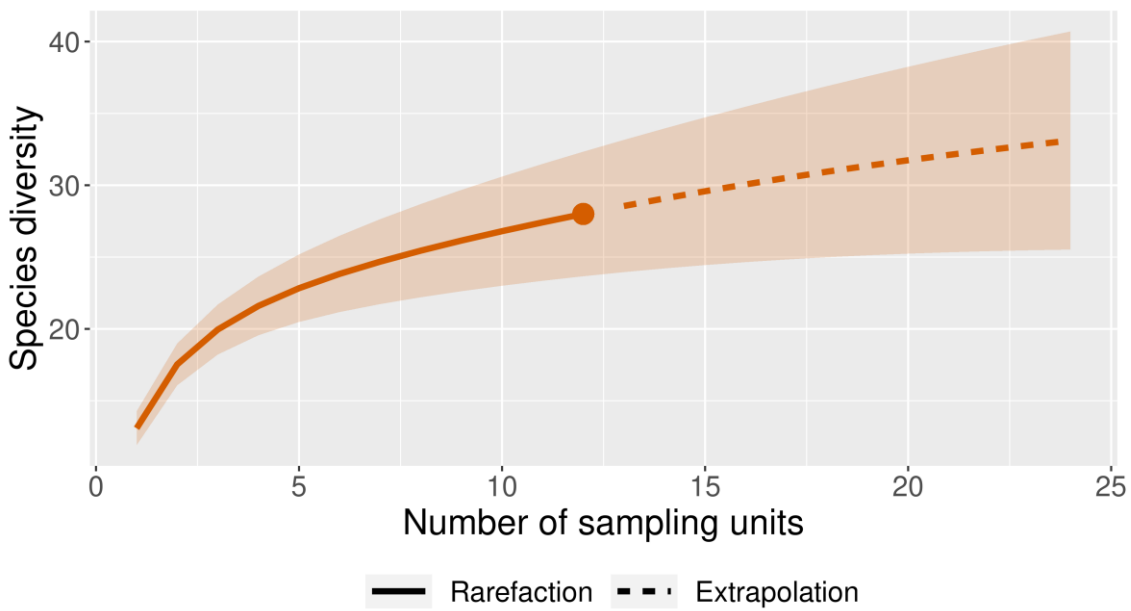
Selective naming at different biological resolutions suggests a classification logic worth testing.

What might underlie the names?

Morphology	shape • size • colour • scales • fins
Ecology	habitat • seasonality • occurrence
Behaviour	where and when fish are encountered, feeding guild, breeding strategy
Harvesting	gear • fishing skill • catchability
Cultural value	food quality • taste • customary use

These are candidate dimensions to investigate with Iban fishers; the present study demonstrates nomenclatural correspondence but does not yet test the diagnostic rules directly.

Citizen science also produced a credible biodiversity inventory



What the coverage means

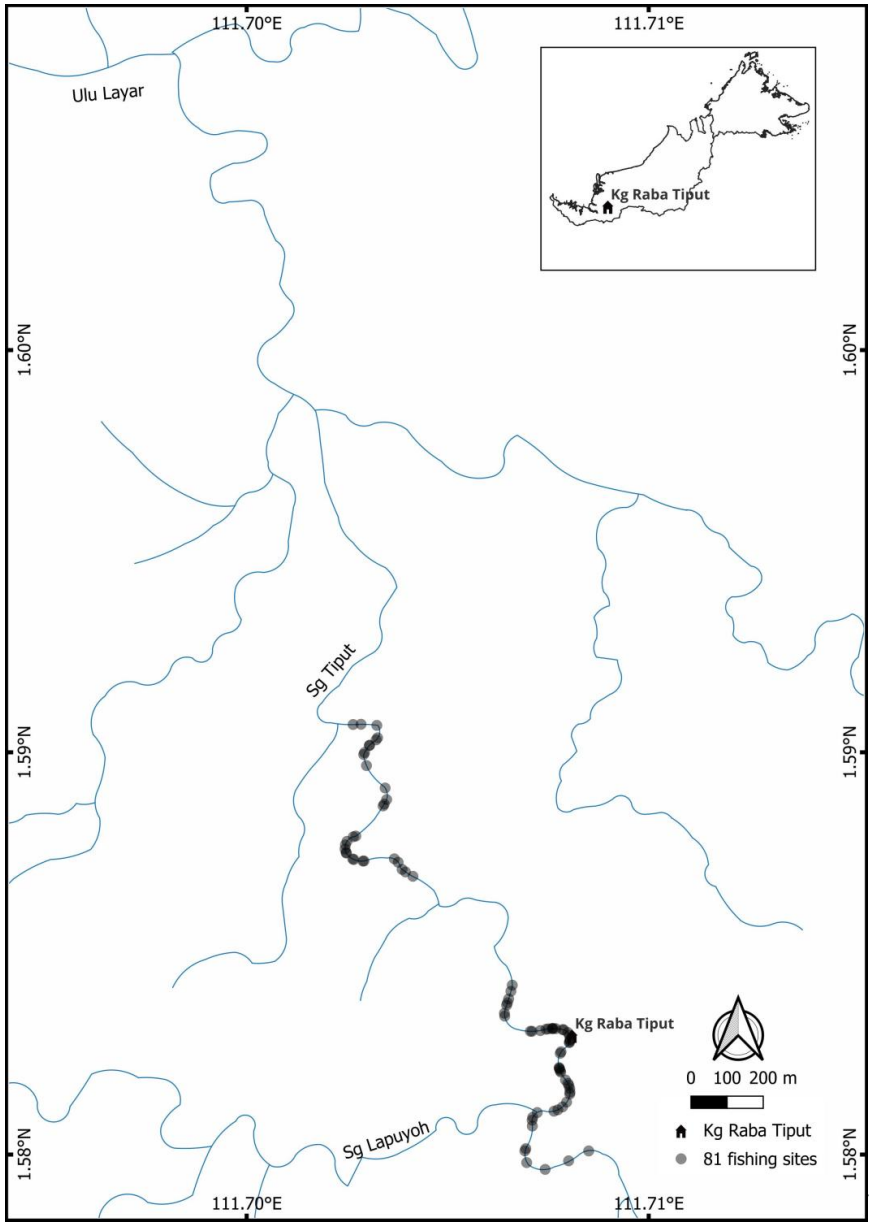
The survey captured nearly all common species; remaining undetected fishes are likely uncommon, cryptic or poorly sampled by the gears used.

What remains open

Richness had not reached an asymptote, so additional effort and broader gears could still add species.

≈96%

sample coverage



What makes this Open Science?

ACCESS

Local participation expands who can contribute observations.

VALIDATION

Taxonomic verification makes observations scientifically interpretable.

RECIPROCITY

Scientific taxonomy and community knowledge are brought into dialogue—not treated as a one-way validation exercise.

REUSE

Validated records can support monitoring, databases and conservation.

Open Science here is not only “open data.” It is opening the process of biodiversity knowledge production.

A caution: openness must respect context and ownership

Do not universalize

The nomenclatural correspondence documented here is specific to Ulu Layar. Iban names can vary geographically and linguistically.



Kejampa – Ulu Layar
Geregit - Julau
Glyptothorax

Do not reduce knowledge to "accuracy"

Species-level mismatch may reflect a broader folk taxon rather than an identification failure.

Do not separate data from people

Prior informed consent, attribution, reciprocity and community control should remain part of the research design.

The bridge must work in both directions.

Next question: can we demonstrate a folk taxonomy?

1. **Document diagnostic cues** — Ask fishers which characters, habitats, behaviours and life stages distinguish each named fish and broader fish group.
2. **Build a bilingual reference** — Link Iban names, scientific names, photographs, vouchers, DNA barcodes where available, and validated records.
3. **Map geographic variation** — Compare nomenclature among Iban communities without assuming one universal naming system.
4. **Strengthen community monitoring** — Support long-term, locally owned biodiversity observations and intergenerational knowledge transfer.

Move from nomenclatural correspondence to documenting the Iban classification system

Conclusion

Iban fish nomenclature in Ulu Layar appears to be more than a set of labels



- The strong genus-level correspondence, species-specific names and broader folk groupings provide evidence consistent with an underlying biologically coherent folk classification system.

Indigenous Knowledge | Scientific Validation | Knowledge Conservation



- Indigenous knowledge and scientific taxonomy need not compete. Through citizen science, they can complement one another in understanding and conserving biodiversity. This aligns with **Article 8(j) of the Convention on Biological Diversity**, which recognises the importance of respecting, preserving and maintaining indigenous knowledge relevant to biodiversity conservation.

Acknowledgements

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Voices of the River









